DevOps Internship — Task 1

Automate Code Deployment Using CI/CD Pipeline (GitHub Actions)

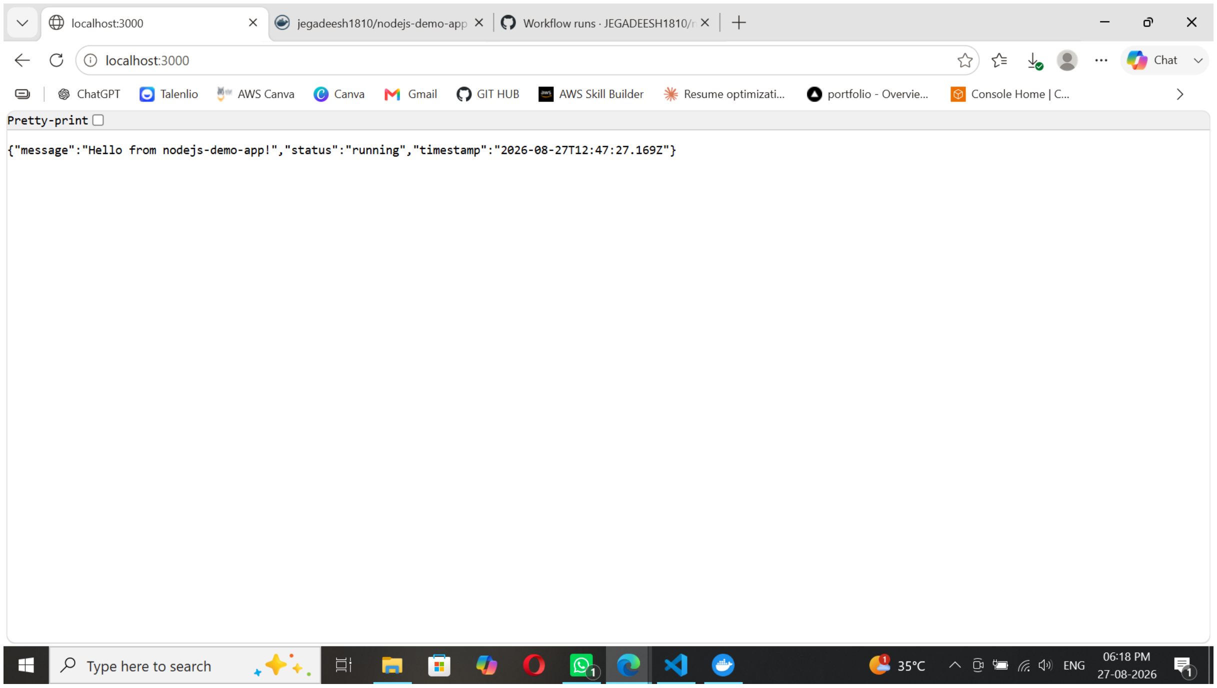
Step-by-Step Evidence Document

Submitted by	JEGADEESH1810
GitHub Repository	github.com/JEGADEESH1810/nodejs-demo-app
Docker Hub Image	hub.docker.com/r/jegadeesh1810/nodejs-demo-app
Tools Used	GitHub, GitHub Actions, Node.js, Docker, Docker Hub

This document presents step-by-step visual evidence that the CI/CD pipeline was built and executed successfully — from the sample Node.js application, through containerization with Docker, to full automation via GitHub Actions (test → build → push to Docker Hub).

Step 1: Sample Node.js Application Running Locally

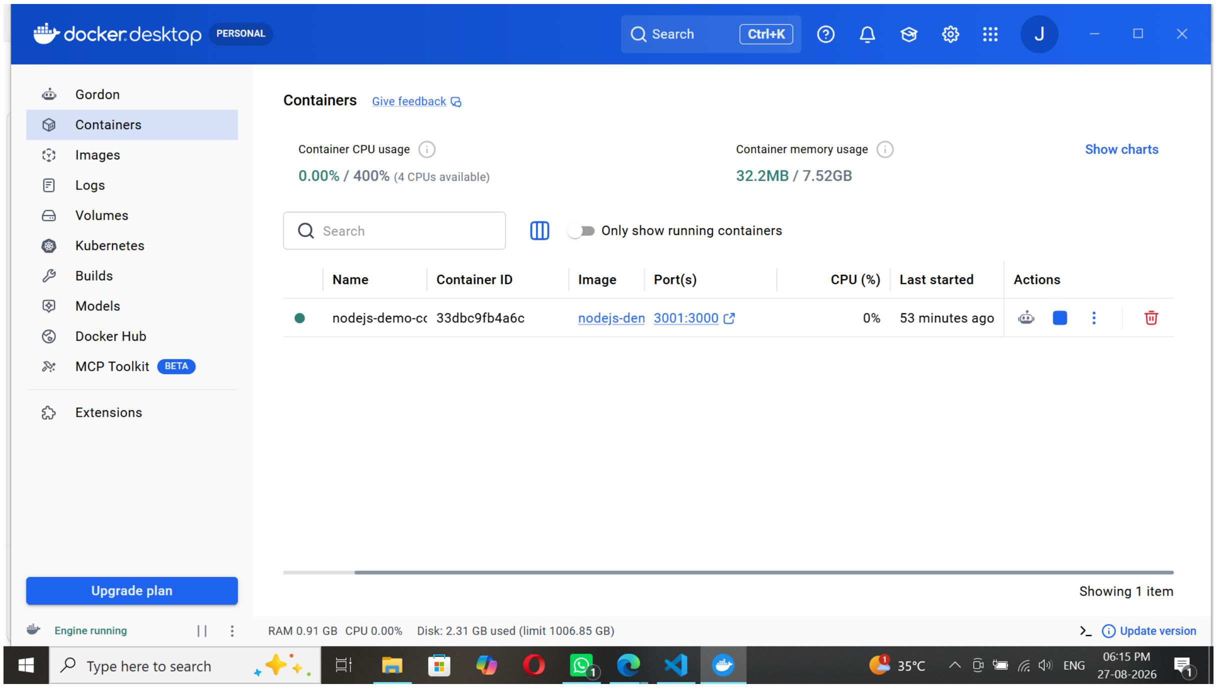
The Express web app was started locally with **npm start** and is accessible at **http://localhost:3000**. The JSON response confirms the app is live and responding correctly.



✓ **This proves:** Sample Node.js app — working and running.

Step 2: Docker Container Running Successfully

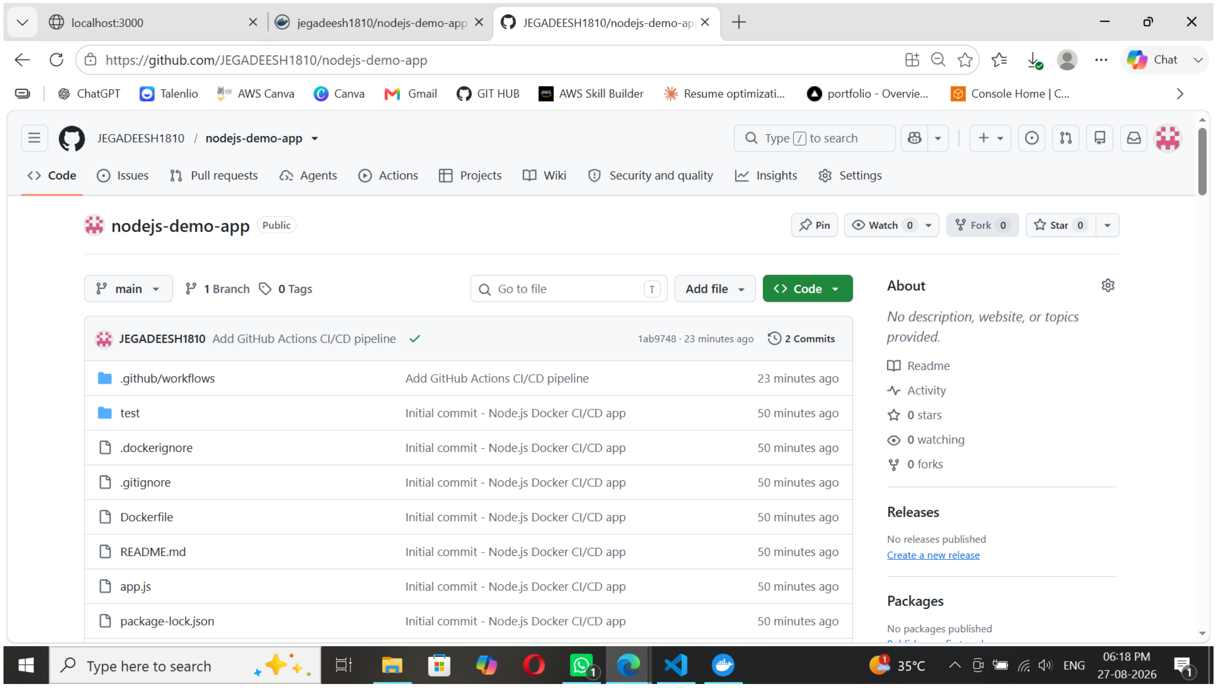
The application was containerized using the project's **Dockerfile** and launched via Docker Desktop. The container **nodejs-demo-container** is shown running (green status), with port **3001** mapped to the container's internal port **3000**.



✓ **This proves:** Docker image built and container running correctly.

Step 3: GitHub Repository — Project Files Pushed

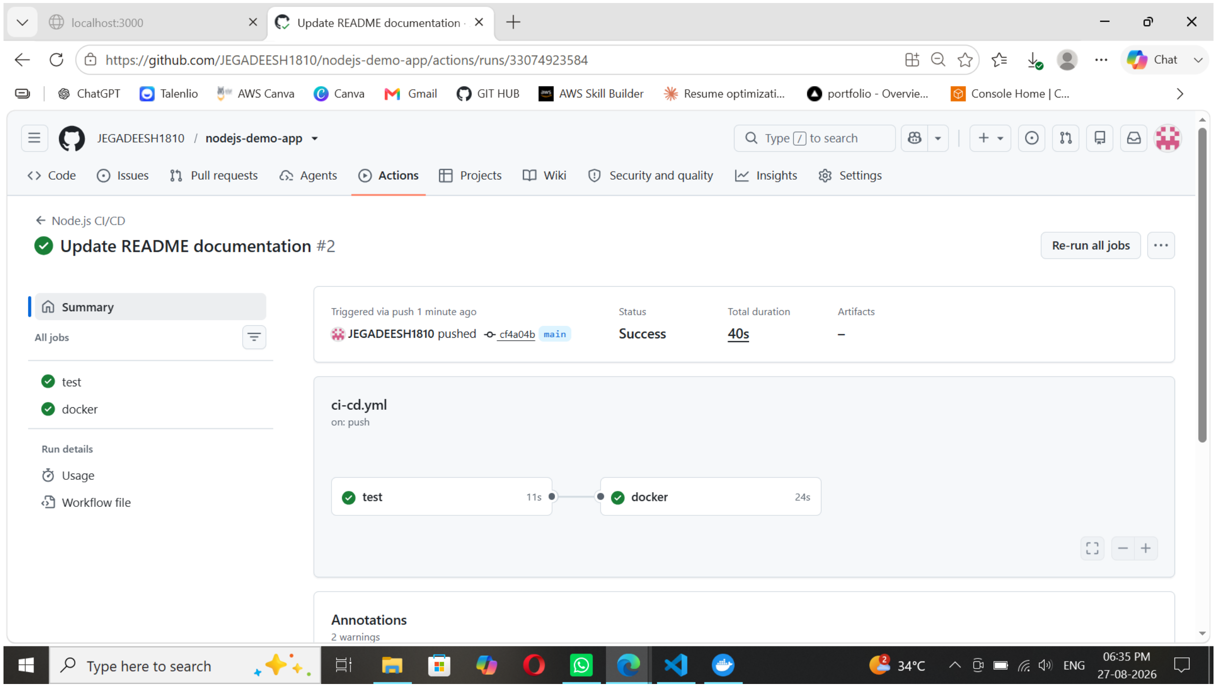
All project deliverables were pushed to the GitHub repository **JEGADEESH1810/nodejs-demo-app**, including the application code, **Dockerfile**, test files, **README.md**, and the **.github/workflows** folder containing the CI/CD pipeline definition.



✓ **This proves:** GitHub repository created with all required files.

Step 4: GitHub Actions — CI/CD Pipeline Executed Successfully

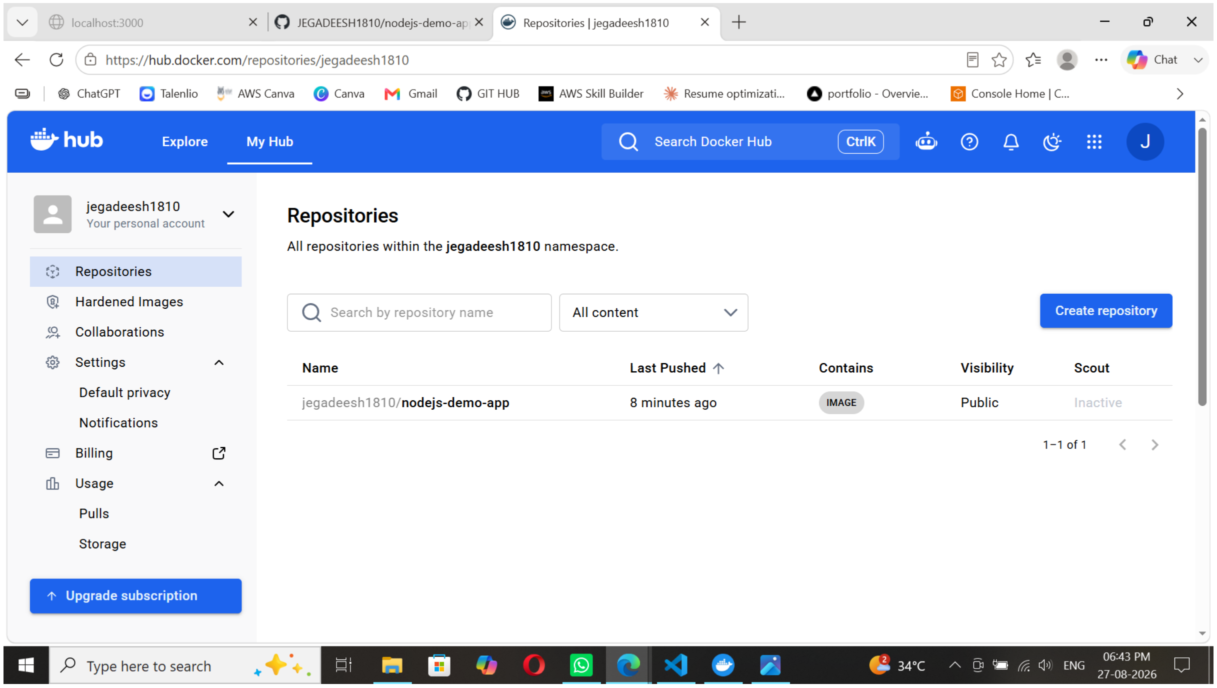
On every push to the **main** branch, the GitHub Actions workflow (**ci-cd.yml**) triggers automatically. Both the **test** job (runs automated tests) and the **docker** job (builds and pushes the Docker image) completed successfully, shown by the green checkmarks and overall **Success** status.



✓ **This proves:** Full automation: test → build → push, triggered on push to main.

Step 5: Docker Hub — Image Successfully Deployed

As the final step of the pipeline, the Docker image was pushed to Docker Hub under the repository **jegadeesh1810/nodejs-demo-app**. The repository is listed as **Public** and shows a recent **Last Pushed** timestamp, confirming a successful automated deployment.



✓ **This proves:** Docker image deployed to Docker Hub via the automated pipeline.

Summary — Task 1 Completion Checklist

Requirement	Status
Sample Node.js application	**Completed**
Dockerfile created	**Completed**
Docker image build & run	**Completed**
GitHub repository created	**Completed**
.github/workflows/ci-cd.yml pipeline	**Completed**
Automated test step	**Completed**
Docker build & push automation	**Completed**
Trigger on push to main	**Completed**
Docker Hub image deployment	**Completed**
README documentation	**Completed**

Repository: github.com/JEGADEESH1810/nodejs-demo-app
Docker Hub: hub.docker.com/r/jegadeesh1810/nodejs-demo-app